



3D printed lock pick

3D printed slices of plastic or metal to form skeleton keys can pick high-security locks. <http://dgit.in/3DPrintKey>



Spaceship Comparison

At Deviantart, a comparison chart has been created with all the sci-fi spaceships and real ones too. <http://dgit.in/SpaceshipChart>

Space Age

ment, the bombardment of radiation can very well turn all potential plant life inert. Given just these basic handful of problems, the task of developing astro-agriculture is a search into the unknown but it is a search we have already begun.

Astroecologist of the Galaxy

The historic research in space agriculture has proven that growing food is a possibility off-planet but demands very high control over the conditions. The International Space Station has been the key setting for these tests and have shown promise on the front. Using the American and Russian developed Lada hardware (Lada is the ancient Russian goddess of spring) aboard the International Space Station, scientists

may arise out of such plant varieties such as change in their genetic make up due to radiation and impact on the space station itself due to environmental adaptation. Mitigating research such as using LED lights to provide specific radiational wavelengths is also being considered.

Since past experiments have shown certain risk factors such as the stress effects of low-gravity environments on plant genetics and appearance, the Lada hardware is critical to this experiment. As this project develops, the goal for astronauts aboard the space station is to grow edible food of their own. This would be an exponential leap towards a feasible long-term space mission which would be self-sustaining with respect to food.

limited finite resources within a closed system, every drop of water from the toilet to the shower is essential. GreenHab systems researches ways to purify and filter out water contaminants which could jeopardise any space mission.

The facility plans to grow high nutrient vegetables in the most limited and resource negligent environments. By growing hardy carrots, radishes, tomatoes and other vegetables along with hydroponic based plants, GreenHab is the testing lab for new varieties and systems of agriculture that can be used on other planets. Independent research programs are also proving to be excellent grounds for space based farming. The most important of which is astroecologist and chemistry professor Dr. Michael Mautner's (Ph.D.) method to grow asparagus, certain fungi and potatoes on meteorite materials. By leveraging the materials abundant in meteors and asteroids such as nitrates, phosphates and water, his method of melding plant growth with extreme-environment microbes growth within a space environment could prove critical to any future ideas for terraforming environments.



While Russians on the ISS sample their grown vegetables, the US harvest is sent to Earth labs for microbe testing.

have observed and studied nearly 500 different plant species. Their most promising project – Validating Vegetable Production Unit Plants, Protocols, Procedures and Requirements Using Existing Flight Resources is currently underway with the goal of optimising the infrastructural hardware and agricultural methodologies aboard the space station.

The project will hopefully identify the varieties of vegetables and flowering plants that are most suited for nutritional and environmental needs. The experiment also takes a very close look at any risks that

Other Earth-bound experiments that seek to address the same challenges of space farming are being conducted in greenhouses such as the GreenHab project. The GreenHab project is a large cylindrical greenhouse 5 metres in height and length, being developed as a part of the Mars Desert Research Station. The facility is being used primarily for the purposes of developing a Mars based greenhouse and a grey water processing unit. Grey water processing is the means by which scientists hope to recycle and reuse waste water that is created. In a space environment, with

"I am zucchini - and I am in space"

As NASA plans its mission to Mars by the mid-2030s, other techniques such as artificial nutritional supplements and even 3D printed food is being explored. Although both these ideas have proven to be effective on Earth they may be insufficient in outer space. The benefits of grown food are not simply a matter of utilitarian use for energy in the human body but also essential for the growth of comfort – both psychological and physical called horticulture therapy. A fun and interesting method of documenting these factors was witnessed in the 2012 blog – "Diary of a Space Zucchini" by astronaut and space-farmer Don Pettit.

Food in whatever form has over thousands of years become a part of human culture which holds people together in a sense of community and family. From the farmers who till the soil to the chefs who make the cuisine – food unifies all people in an ephemeral manner and even though people like to say armies march on their stomachs, its truer still that humanity dreams on a full stomach. Let's get growing. We are Groot! 